**8** **(a)** State the *principle of conservation of momentum*.

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**(b)** In a nuclear reactor, uranium-235 decays to produce two or three neutrons and a few daughter nuclei. For the chain reaction to continue, the speed of the neutrons that produced from the decay of the uranium-235 need to be slowed down sufficiently before colliding with another uranium-235. Carbon atom is commonly used to slow down the neutrons.

**(i)** A neutron of mass $m$ and speed $u$ collides elastically head-on with a stationary carbon atom of mass $M$. The speed of the neutron and the carbon atom after the collision are $v$ and $V$ respectively.

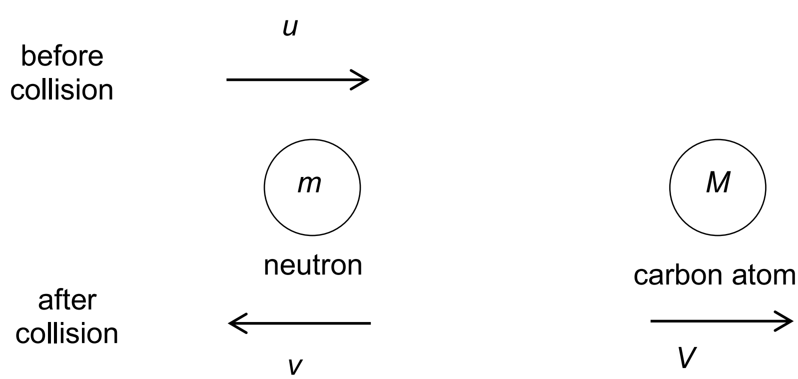


**Fig. 8.1**

**1.** Using Newton's Laws of Motion, show that the principle of conservation of linear momentum for the neutron and the carbon atom is

$$mu = MV - mv$$

Explain your working clearly.

[3]

**2.** Given that $v$ and $u$ is related by the expression,

$$v = \left(\frac{M - m}{M + m}\right)u$$

determine the fraction of kinetic energy retained by the neutron after it collides with a carbon atom with mass $M$ of $12u$.
The mass of a neutron is $1.67 \times 10^{-27}$ kg.

fraction = .............................. [2]

**3.** Hence, calculate the number of such head-on collisions needed to reduce the kinetic energy of the neutron to one millionth of its original value.

number of collisions = ............................. [1]

**4.** Explain why very heavy elements are unsuitable for slowing down neutrons.

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**(ii)** After the neutron has been slowed down sufficiently, it collided with another uranium-235 atom.

Explain whether this collision is elastic.

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**(c)** A scientist fired a bullet at a 1.0 kg block that is suspended by two 15.0 cm thin threads from a fixed point. The bullet is embedded in the block, which swings upwards to a maximum vertical displacement of 3.6 cm from its original position as shown in Fig 8.2.

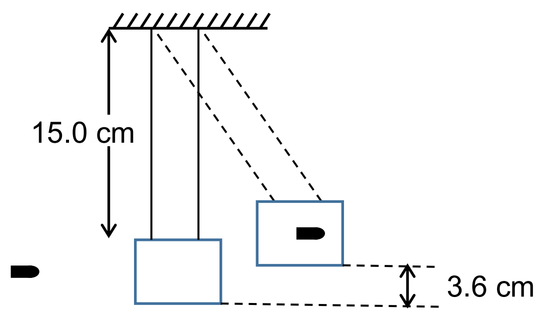


**Fig. 8.2**

**(i)** Describe the energy changes as the bullet approaches the block until the time the block and bullet reaches the highest point.

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**(ii)** Show that the speed of the block and bullet just after the impact by the bullet is 0.84 m $s^{-1}$.

[1]

**(iii)** The bullet has a mass of 2.6 g.

Calculate the speed of the bullet just before impact.

speed of bullet = ........................... m $s^{-1}$ [2]

**(iv)** The bullet was found to be lodged 1.4 cm into the block.

Determine the average frictional force exerted on the bullet.

average frictional force = ........................ N [3]